

Project Documentation

File: line simulation v2.project

Date: 03/07/2026

Profile: CODESYS V3.5 SP20 Patch 1

1 DUT: e_state

```
1  TYPE e_state :
2  (
3      OFF := 1 , // LINE IS OFF
4      HEATING := 2 , // OVEN IS HEATING, HEATING ELEMENT IS ON
5      WORKING := 3 , // THE OVEN IS IN WORKING TEMPTURE
6      UNLOADING := 4 , // THE LINE IS CLEARING PRODUCT
7      COOLING := 5 , // THE OVEN IS OVER HEATING HEATING ELEMENT IS OFF
8      CLEARING := 6 , //CLEARING PRODUCT FROM LINE
9      EMERGENCY := 7 // THE LINE IS STOPED
10 ) ;
11 END_TYPE
12
```

2 Global Variable List: GVL

```
1  {attribute 'qualified_only'}
2  VAR_GLOBAL
3      iRawAnalogIn : INT ; // GLOBAL ANALOG INPUT OF RAW TEMPTURE
4      rTempCout : REAL ; // GLOBAL TEMPTURE SENSOR OUTPUT IN CELCIUS
5      bLocationsensor : BOOL ; // GLOBAL PRUDUCT LOCATION SENSOR
6  END_VAR
7
```

3 POU: fb_OvenState

```
1  FUNCTION_BLOCK fb_OvenState
2  VAR_INPUT
3      bSTART : BOOL ; // START LINE OPERATION
4      bSTOP : BOOL ; // STOP LINE OPERATION
5      bRESET : BOOL ; // RESET BUTTON TO CLEAR FAULTS AND EMERGENCY STATES
6      rCorentTemp : REAL ; // CURRENT TEMPERATURE FEEDBACK FROM OVEN SENSOR
7      rTargetTemp : REAL := 200.0 ; // TARGET WORKING TEMPERATURE SETPOINT
8      bProductLocation : BOOL ; // PRODUCT DETECTION SENSOR INSIDE THE OVEN
9      rRecordFbTemp : REAL ; // INPUT OF THE LAST RECORDED TEMP FROM THE
10 RECORD_BLOCK
11 END_VAR
12 VAR_OUTPUT
13     eWorkstate : e_state ; // CURRENT MACHINE STATE BASED ON E_STATE ENUM
14     bHeatingElemnt : BOOL ; // HEATING ELEMENT CONTROL COMMAND (ON/OFF)
15     bFan : BOOL ; // OVEN COOLING FAN CONTROL COMMAND (ON/OFF)
16     bConveyorF : BOOL ; // CONVEYOR FORWARD MOVEMENT COMMAND
17     bConveyorB : BOOL ; // CONVEYOR BACKWARD MOVEMENT COMMAND
18     sMessage : STRING ; // SYSTEM STATUS MESSAGE FOR HMI DISPLAY
19 END_VAR
20     tHeatCheckTimer : TON ; // TIMER FOR CHECKING OVEN TEMPERATURE
21     tProsesTimer : TON ; // TIMER FOR PRODUCT PROCESSING DURATION
22     tOvenHeatingTimer : TON ; // HEATING TIMEOUT MONITORING TIMER
23     tTempCheckTimer : TON ; // TIMER FOR CHECKING IF COOLING IS WORKING
```

3 POU: fb_OvenState

```
24
25     // INTERNAL CONTROL FLAGS TO PROPERLY TRIGGER AND RESET TIMERS OUTSIDE
    THE CASE STRUCTURE
26     bTriggerHeatCheck : BOOL ;
27     bTriggerProses : BOOL ;
28     bTriggerOvenHeating : BOOL ;
29     bTriggerTempCheck : BOOL ; // TRIGGER FLAG FOR THE COOLING CHECK TIMER
30
31     // INTERNAL CONTROL FLAGS FOR SAFETY AND PRODUCT SWAP MANAGEMENT
32     bSafetyOk : BOOL ; // SAFETY INTERLOCK FLAG (TRUE WHEN SYSTEM IS SAFE
    TO RUN)
33     bProductUnloading : BOOL ; // HANDSHAKE FLAG TO TRACK COMPLETED PRODUCT
    DISCHARGE
34 END_VAR
35
```

```
1 // SAFETY INTERLOCK & RESET LOGIC
2 // IF STOP BUTTON IS PRESSED, SAFETY CLEARANCE IS LOST.
3 // RESET IS ONLY ALLOWED IF UNDER CATASTROPHIC TEMPERATURE LIMIT.
4 IF bSTOP THEN
5     bSafetyOk := FALSE ;
6 ELSIF bRESET AND ( rCorentTemp <= ( rTargetTemp + 30.0 ) ) THEN
7     bSafetyOk := TRUE ;
8 END_IF
9
10 // FORCE THE MACHINE TO THE OFF STATE IF SAFETY IS LOST AND NOT IN A SAFE
    HANDLING STATE
11 IF NOT bSafetyOk AND eWorkstate <> e_state.OFF AND eWorkstate <>
    e_state.CLEARING AND eWorkstate <> e_state.COOLING AND eWorkstate <>
    e_state.EMERGENCY THEN
12     eWorkstate := e_state.OFF ;
13 END_IF
14
15 // MAIN STATE MACHINE LOGIC
16 CASE eWorkstate OF
17
18     e_state.OFF :
19         // TURN OFF ALL PHYSICAL COMPONENTS WHEN THE SYSTEM IS INACTIVE
20         bHeatingElemnt := FALSE ;
21         bFan := FALSE ;
22         bConveyorB := FALSE ;
23         bConveyorF := FALSE ;
24         sMessage := 'SYSTEM IS OFF - PRESS RESET THEN START TO
    ACTIVATE' ;
25
26         // RESET ALL OPERATIONAL FLAGS AND TIMERS TO CLEAR MEMORY FROM
    PREVIOUS RUNS
27         bTriggerHeatCheck := FALSE ;
28         bTriggerOvenHeating := FALSE ;
29         bTriggerTempCheck := FALSE ;
30         bTriggerProses := FALSE ;
31         bProductUnloading := FALSE ;
32
33         // PRIORITY 1: IF A PRODUCT IS LEFT INSIDE ON STARTUP, GO TO
    CLEARING STATE IMMEDIATELY
```

3 POU: fb_OvenState

```
34         IF bProductLocation = TRUE THEN
35             eWorkstate := e_state . CLEARING ;
36
37             // PRIORITY 2: ENVIRONMENTAL SAFETY CHECK - ONLY FORCE COOLING IF
OVEN IS ACTUALLY TOO HOT ABOVE ROOM TEMP
38             ELSIF ( rCorentTemp > ( rTargetTemp + 15.0 ) ) AND ( rCorentTemp >
50.0 ) THEN
39                 sMessage := 'OVEN TOO HOT! AUTOMATIC COOLING ACTIVATED' ;
40                 eWorkstate := e_state . COOLING ;
41
42                 // PRIORITY 3: START SEQUENCE
43             ELSIF bSTART AND bSafetyOk THEN
44                 eWorkstate := e_state . HEATING ;
45             END_IF
46
47         e_state . HEATING :
48             bHeatingElemnt := TRUE ;
49             bFan := FALSE ;
50             bConveyorB := FALSE ;
51             bConveyorF := FALSE ;
52             sMessage := 'OVEN HEATING UP' ;
53
54             bTriggerHeatCheck := TRUE ;
55             bTriggerOvenHeating := TRUE ;
56
57             // TREND PROTECTION: ONLY CHECK TREND IF OVEN HAS PASSED ROOM
TEMPERATURE INITIALIZATION
58             IF tHeatCheckTimer . Q AND rCorentTemp > 50.0 THEN
59                 IF rCorentTemp <= rRecordFbTemp THEN
60                     bTriggerOvenHeating := FALSE ;
61                     bTriggerHeatCheck := FALSE ;
62                     bSafetyOk := FALSE ;
63                     sMessage := 'HEATING FAILURE! OVEN TEMPERATURE
IS NOT RISING' ;
64                     eWorkstate := e_state . EMERGENCY ;
65                 END_IF
66             END_IF
67
68             // IF THE HEATING TIMER FINISHES ITS COUNTDOWN AND THE OVEN IS STILL
COLD
69             IF tOvenHeatingTimer . Q AND ( rCorentTemp < rTargetTemp ) THEN
70                 bTriggerOvenHeating := FALSE ;
71                 bTriggerHeatCheck := FALSE ;
72                 sMessage := 'HEATING READ TIME OUT! EMERGENCY STATE'
;
73                 eWorkstate := e_state . EMERGENCY ;
74             END_IF
75
76             // IF CURRENT TEMPERATURE REACHES OR EXCEEDS THE TARGET TEMPERATURE
77             IF rCorentTemp >= rTargetTemp THEN
78                 bTriggerOvenHeating := FALSE ;
79                 bTriggerHeatCheck := FALSE ;
80                 eWorkstate := e_state . WORKING ;
81             END_IF
82
83         e_state . WORKING :
```

3 POU: fb_OvenState

```
84      // THERMAL RUNAWAY PROTECTION
85      IF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
86          bTriggerProses := FALSE ;
87          eWorkstate      := e_state . COOLING ;
88      END_IF
89
90      // FIXED HARDWARE FAULT PROTECTION (CRASH DETECTION)
91      // Lowered to 40.0 to prevent immediate deadlock at room temp during
startup simulation
92      IF rCorentTemp < 40.0 THEN
93          bTriggerProses := FALSE ;
94          bSafetyOk      := FALSE ;
95          sMessage       := 'CRITICAL SENSOR OR ELEMENT FAILURE DETECTED!
IMPLAUSIBLE VALUE' ;
96          eWorkstate     := e_state . EMERGENCY ;
97      END_IF
98
99      // PROCEED WITH NORMAL LOGIC ONLY IF WE ARE NOT IN A HARDWARE
EMERGENCY TRAP
100     IF eWorkstate = e_state . WORKING THEN
101         IF bProductLocation = TRUE THEN
102             bConveyorF      := FALSE ;
103             bTriggerProses   := TRUE ;
104             sMessage        := 'PRODUCT IN PROCESS' ;
105             bHeatingElemnt   := ( rCorentTemp < rTargetTemp ) ;
106
107             IF tProsesTimer . Q THEN
108                 bTriggerProses := FALSE ;
109                 IF ( rCorentTemp >= ( rTargetTemp - 15.0 ) ) AND (
rCorentTemp <= ( rTargetTemp + 15.0 ) ) THEN
110                     sMessage      := 'PROCESS SUCCESSFUL - ROUTING
TO UNLOAD' ;
111                     bProductUnloading := TRUE ;
112                     eWorkstate     := e_state . UNLOADING ;
113                 ELSE
114                     sMessage      := 'PROCESS TEMPERATURE FAULT! REJECTING
ITEM' ;
115                     eWorkstate := e_state . CLEARING ;
116                 END_IF
117             END_IF
118         ELSE
119             IF NOT bTriggerProses THEN
120                 bConveyorF      := TRUE ;
121                 bHeatingElemnt   := ( rCorentTemp < rTargetTemp ) ; //
Maintain stable chamber heat while waiting
122                 sMessage        := 'LOADING NEW PRODUCT' ;
123             END_IF
124         END_IF
125     END_IF
126
127     e_state . UNLOADING :
128     bConveyorF      := TRUE ;
129     bHeatingElemnt   := ( rCorentTemp < rTargetTemp ) ;
130     bTriggerProses   := FALSE ;
131     sMessage         := 'PRODUCT SWAP IN PROGRESS - WAITING FOR ENTRY' ;
132
```

3 POU: fb_OvenState

```
133         IF NOT bProductLocation THEN
134             bProductUnloading := FALSE ;
135             sMessage           := 'PREVIOUS PRODUCT CLEARED - WAITING FOR
NEXT' ;
136         END_IF
137
138         IF NOT bProductUnloading AND bProductLocation = TRUE THEN
139             sMessage := 'NEW PRODUCT DETECTED - RESTARTING PROCESS' ;
140             eWorkstate := e_state . WORKING ;
141         END_IF
142
143         IF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
144             eWorkstate := e_state . COOLING ;
145         END_IF
146
147         e_state . COOLING :
148             bHeatingElemnt := FALSE ;
149             bFan            := TRUE ;
150             bConveyorF     := FALSE ;
151             bTriggerTempCheck := TRUE ;
152             sMessage       := 'COOLING IN PROGRESS, REJECTING PRODUCT' ;
153
154             IF bProductLocation = TRUE THEN
155                 bConveyorB := TRUE ;
156             ELSE
157                 bConveyorB := FALSE ;
158             END_IF
159
160             // TREND PROTECTION FOR COOLING
161             IF tTempCheckTimer.Q AND rCorentTemp > 50.0 THEN
162                 IF rCorentTemp >= rRecordFbTemp THEN
163                     bTriggerTempCheck := FALSE ;
164                     bSafetyOk         := FALSE ;
165                     sMessage          := 'COOLING FAILURE DETECTED! OVEN
TEMPERATURE IS NOT DROPPING' ;
166                     eWorkstate       := e_state . EMERGENCY ;
167                 END_IF
168             END_IF
169
170             // CATASTROPHIC ESCALATION CHECK & EXIT CONDITIONS
171             IF eWorkstate = e_state . COOLING THEN
172                 IF ( rCorentTemp > ( rTargetTemp + 30.0 ) ) AND NOT bRESET AND
NOT bSTOP THEN
173                     bTriggerTempCheck := FALSE ;
174                     bSafetyOk         := FALSE ;
175                     sMessage          := 'CRITICAL THERMAL ESCALATION! HARD
LOCKED' ;
176                     eWorkstate       := e_state . EMERGENCY ;
177
178                     ELSIF rCorentTemp > ( rTargetTemp + 15.0 ) THEN
179                         sMessage := 'CRITICAL COOLING ACTIVE' ;
180
181                     ELSIF rCorentTemp <= rTargetTemp THEN
182                         // FIX: If oven cooled down and empty -> OFF. If still
contains product -> CLEARING
183                         bTriggerTempCheck := FALSE ;
```

3 POU: fb_OvenState

```
184         IF bProductLocation = FALSE THEN
185             eWorkstate := e_state.OFF ;
186         ELSE
187             eWorkstate := e_state.CLEARING ;
188         END_IF
189     END_IF
190 END_IF
191
192 e_state.CLEARING :
193     bHeatingElemnt := FALSE ;
194     bFan             := FALSE ;
195     bConveyorF       := FALSE ;
196     sMessage         := 'CLEARING PRODUCT FROM LINE' ;
197
198     IF bProductLocation = TRUE THEN
199         bConveyorB := TRUE ;
200     ELSE
201         bConveyorB := FALSE ;
202         eWorkstate := e_state.OFF ;
203     END_IF
204
205 e_state.EMERGENCY :
206     bHeatingElemnt := FALSE ;
207     bFan             := FALSE ;
208     bConveyorB       := FALSE ;
209     bConveyorF       := FALSE ;
210     sMessage         := 'OVEN FAILURE! CHECK HEATING ELEMENTS AND SENSOR'
211 ;
212
213     // FIX: Route to COOLING only if temperature is actually high,
214     // otherwise route directly to OFF
215     IF bSTOP OR bRESET THEN
216         IF rCorentTemp > rTargetTemp THEN
217             eWorkstate := e_state.COOLING ;
218         ELSE
219             bSafetyOk := TRUE ; // Auto-clear safety block to allow
220             // initialization restart
221             eWorkstate := e_state.OFF ;
222         END_IF
223     END_IF
224
225 END_CASE
226
227 // TIMERS EXECUTION CALLS
228
229 tHeatCheckTimer ( IN := bTriggerHeatCheck , PT := T#5S ) ;
230 tOvenHeatingTimer ( IN := bTriggerOvenHeating , PT := T#15S ) ;
231 tTempCheckTimer ( IN := bTriggerTempCheck , PT := T#20S ) ;
232 tProsesTimer ( IN := bTriggerProses , PT := T#10S ) ;
233
234 // INTERNAL OVEN SIMULATION LOGIC (SIMULATION ONLY)
235
236 // Simulates live physical feedback so internal safety timers do not crash
237 // during pure simulation
```

3 POU: fb_OvenState

```
236     IF bHeatingElemnt THEN
237         rCorentTemp := rCorentTemp + 0.15 ;
238     ELSIF bFan THEN
239         rCorentTemp := rCorentTemp - 0.25 ;
240     ELSE
241         IF rCorentTemp > 25.0 THEN
242             rCorentTemp := rCorentTemp - 0.01 ; // Slow environmental heat loss
243         END_IF
244     END_IF
245     // Automatically adjusts trend feedback references when active to prevent
246     // static value interlock trip
247     IF eWorkstate = e_state.HEATING AND NOT tHeatCheckTimer.Q THEN
248         rRecordFbTemp := rCorentTemp - 1.0 ; // Forces system to register an
249         // active heating curve
250     ELSIF eWorkstate = e_state.COOLING AND NOT tTempCheckTimer.Q THEN
251         rRecordFbTemp := rCorentTemp + 1.0 ; // Forces system to register an
252         // active cooling curve
253     END_IF
```

4 POU: fb_sensorRcored

```
1  FUNCTION_BLOCK fb_sensorRcored
2  VAR_INPUT
3      rCorentLogTemp : REAL ; // CURRENT TEMPERATURE FEEDBACK TO BE LOGGED
4      tLogInterval : TIME := T#20S ; // TIME CYCLE FOR EACH LOG ENTRY
5      (DEFAULT 20 SECONDS)
6  END_VAR
7  VAR_OUTPUT
8      // AN ARRAY STORES THE LAST 10 LOGGED TEMPERATURE VALUES FOR SIMULATION
9      TRACKING
10     aTempHistory : ARRAY [ 1 .. 10 ] OF REAL ;
11 END_VAR
12 VAR
13     tLogTimer : TON ; // INTERNAL TIMER TO TRIGGER THE LOG EVERY 20 SECONDS
14     i : INT ; // LOOP INDEX FOR MOVING ARRAY DATA
15     rLastStoredTemp : REAL ;
16 END_VAR

1 // EXECUTE THE REPEATING LOG TIMER WITH AUTOMATIC RESTART TRIGGER
2 tLogTimer ( IN := NOT tLogTimer.Q , PT := tLogInterval ) ;
3
4 // EVERY TIME THE TIMER REACHES 20 SECONDS, TRIGGER THE SHIFT AND RECORD
5 // SEQUENCE
6 IF tLogTimer.Q THEN
7     // SHIFT ALL EXISTING LOGGED VALUES ONE SLOT DOWN IN THE ARRAY FROM
8     // POSITION 1 UP TO 9
9     FOR i := 1 TO 9 BY 1 DO
10         aTempHistory [ i ] := aTempHistory [ i + 1 ] ;
11     END_FOR ;
```

4 POU: fb_sensorRcored

```
11
12      // INSERT THE LATEST CURRENT TEMPERATURE VALUE INTO THE NEWEST POSITION
      SLOT 10
13      aTempHistory [ 10 ] := rCorentLogTemp ;
14
15      // UPDATE THE OUTPUT WITH THE RECENTLY RECORDED VALUE TO SEND IT TO THE
      OVEN STATE MACHINE
16      rLastStoredTemp := rCorentLogTemp ;
17
18  END_IF
19
20  // INITIALIZATION PROTECTION: IF THE LOGGED TEMP IS STILL ZERO DURING THE
      FIRST 20 SECONDS, CAPTURE CURRENT TEMP AS BASELINE
21  IF rLastStoredTemp = 0.0 THEN
22      rLastStoredTemp := rCorentLogTemp ;
23  END_IF
24
```

5 POU: fb_SensorScaling

```
1  FUNCTION_BLOCK fb_SensorScaling
2  VAR_INPUT
3      X : INT ; // INPUT OF iRawAnalogIn SENSOR TO SCALING FUNCTION
4      X1 : REAL := 0.0 ; // LOWER LIMIT OF THE SENSOR
5      X2 : REAL := 27648.0 ; // UPER LIMIT OF THE SENSOR BASED ON SIEMNES PLC
6      y1 : REAL := 23.0 ; // LOWER OUTPUT RANGE FOR SENSOR TO SIMOLATE ROOM
      TEMPTURE
7      y2 : REAL := 400.0 ; // UPER OUTPOT RANGE FOR SENSOR IN CELCIUS
8  END_VAR
9  VAR_OUTPUT
10     y : REAL ; // OUTPUT OF rTempCout SENSOR FROM SCALING FUNCTION
11 END_VAR
12 VAR
13 END_VAR
14
```

```
1      y := y1 + ( INT_TO_REAL ( x ) - x1 ) * ( y2 - y1 ) / ( x2 - x1 ) ; //
      SCALING FUNCTION FOR SENSOR
2
```

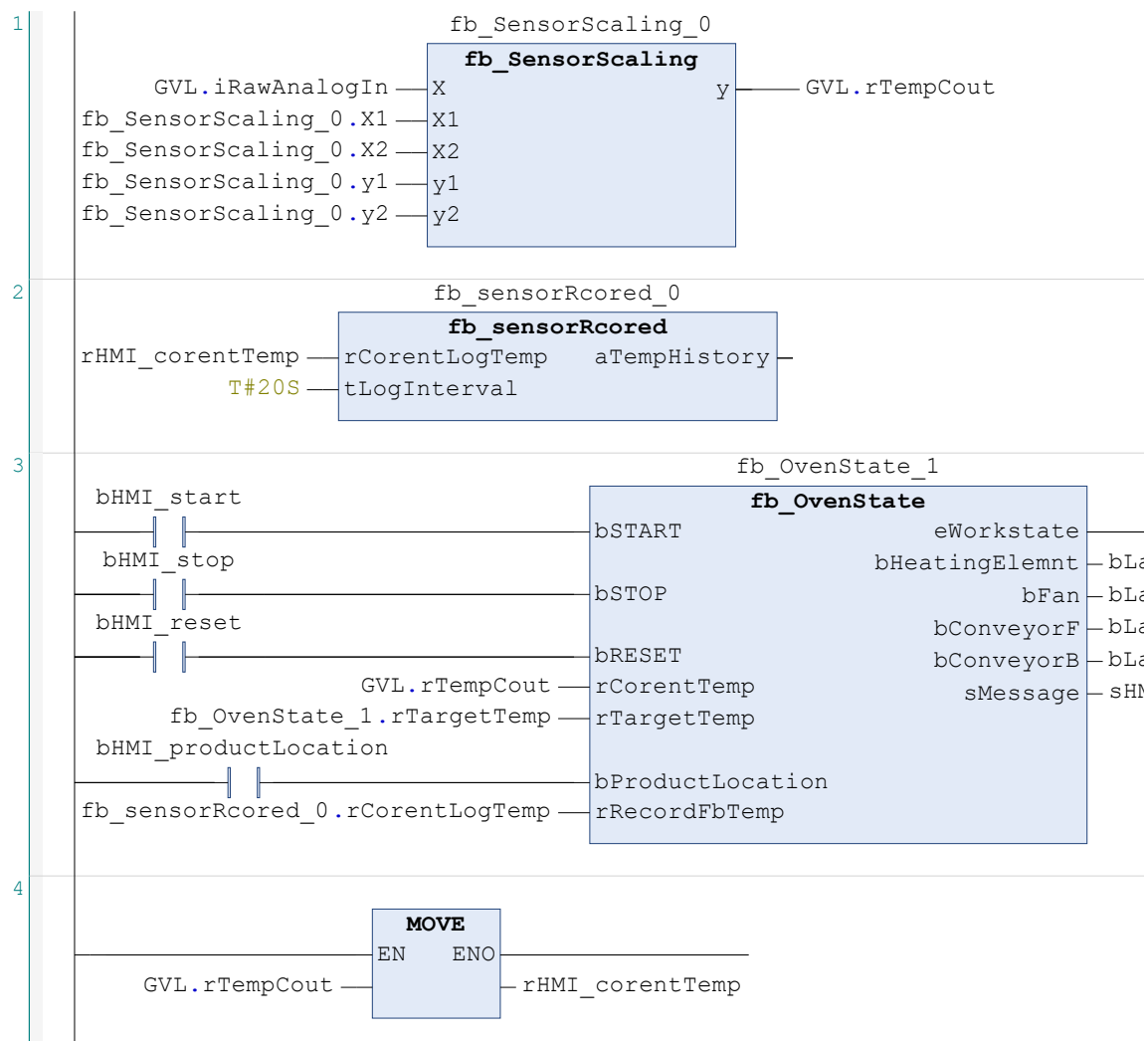
6 POU: PLC_PRG

```
1  PROGRAM PLC_PRG
2  VAR
3      fb_SensorScaling_0 : fb_SensorScaling ; // DEFNATION OF THE TEMTURE
      SCALING FUNCTIONAL BLOCK
4      fb_OvenState_1 : fb_OvenState ; //DEFNATION OF THE OVEN STATE FUNCTUNAL
      BLOCK
5      fb_sensorRcored_0 : fb_sensorRcored ; // DEFNATION OF THE RECOREDING
      SENSOR BLOCK
6      // LINKING INPUTS
7      bHMI_start : BOOL ;
8      bHMI_stop : BOOL ;
9      bHMI_reset : BOOL ; // NEW: HMI RESET BUTTON TO CLEAR EMERGENCIES AND
```

```

10      SAFETY_FAULTS
11      rHMI_corentTemp : REAL ;
12      rHMI_targetTemp : REAL := 200.0 ;
13      bHMI_productLocation : BOOL ;
14
15      // LINKING OUTPUTS
16      bLamp_heating : BOOL ;
17      bLamp_Fan : BOOL ;
18      bLamp_ConveyorF : BOOL ;
19      bLamp_ConveyorB : BOOL ;
20      bLamp_Reset : BOOL ; // NEW: RESET PROMPT LAMP (FLASHES WHEN RESET IS
21      REQUIRED)
22      sHMI_Message : STRING ;
23
24      END_VAR

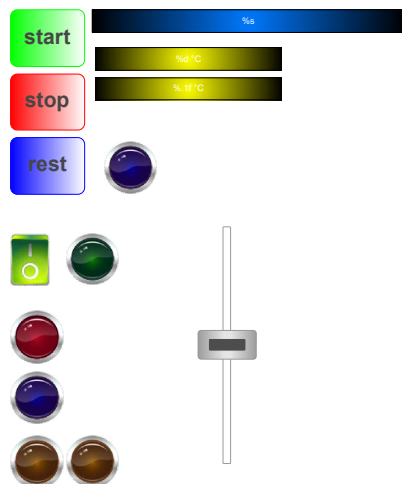
```



amp_heating
amp_Fan
amp_ConveyorF
amp_Conveyorb
4I_Message

7 Program Call: PLC_PRG

8 Visualization: Visualization



General

Visualization size algorithm version: Respecting scrollbar location

Background

Use background color: False

Background color: 16777215

Interface

VAR_IN_OUT

END_VAR

Visualization Element List

Button Id: 0

Element name: GenElemInst_2

Type of element: Button

Tab Order: default

Static optimized: False

Text ID: 467

Position

X: 28

Y: 23

Width: 98

Height: 75

Center

X: 77

Y: 60

Colors

Color

Transparency: 255

Alarm color

Transparency: 255
Use gradient color: True
Gradient setting
 Gradient
 Color1: Green
 Color2: Color [A=255, R=255, G=255, B=255]
 Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
 Scaling type: Isotropic
 Transparent: False
 Transparent color: Black
 Horizontal alignment: Left
 Vertical alignment: Top
Texts
 Text: start
Text properties
 Horizontal alignment: Centered
 Vertical alignment: Centered
 Text format: Default
 Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
 Font color
 Transparency: 255
Font variables
 Size: , <pt>
Button state variable
 Digital variable: <NOT toggle/tap variable>
Animation duration: 0
Input configuration
 OnDialogClosed: Configure...
 OnMouseClicked: Configure...
 OnMouseDown: Configure...
 OnMouseEnter: Configure...
 OnMouseLeave: Configure...
 OnMouseMove: Configure...
 OnMouseUp: Configure...
 OnValueChanged: Configure...
Toggle
 Toggle on up if captured: False
Tap
 Variable: PLC_PRG.bHMI_start
 Tap FALSE: True
 Tap on enter if captured: False
■ Button Id: 3
 Element name: GenElemInst_3
 Type of element: Button
 Tab Order: default
 Static optimized: False
 Text ID: 514
Position
 X: 28
 Y: 108
 Width: 98
 Height: 75

Center
X: 77
Y: 145
Colors
Color
Transparency: 255
Alarm color
Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Red
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
Scaling type: Isotropic
Transparent: False
Transparent color: Black
Horizontal alignment: Left
Vertical alignment: Top
Texts
Text: stop
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
Font color
Transparency: 255
Font variables
Size: , <pt>
Button state variable
Digital variable: <NOT toggle/tap variable>
Animation duration: 0
Input configuration
OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Variable: PLC_PRG.bHMI_stop
Tap FALSE: True
Tap on enter if captured: False
■ RockerSwitch Id: 5
Element name: GenElemInst_5
Type of element: RockerSwitch
Tab Order: default

Static optimized: False
Position
 X: 28
 Y: 319
 Width: 70
 Height: 70
Variable: PLC_PRG.bHMI_productLocation
Image settings
 Transparent: True
 Transparent color: BasicElement-Fill-Color
 Isotropic type: Isotropic
 Horizontal alignment: Left
 Vertical alignment: Top
Element behavior: Image toggler
Tap FALSE: False
Center
 X: 63
 Y: 354
Animation duration: 0
Input configuration
 OnValueChanged: Configure...
Background
 Image: Green

● Lamp1 Id: 7
 Element name: GenElemInst_7
 Type of element: Lamp1
 Tab Order: default
 Static optimized: False
 Position
 X: 28
 Y: 423
 Width: 70
 Height: 70
 Variable: PLC_PRG.bLamp_heating
 Image settings
 Transparent: False
 Transparent color: Red
 Isotropic type: Isotropic
 Horizontal alignment: Left
 Vertical alignment: Top
 Center
 X: 63
 Y: 458
 Animation duration: 0
 Background
 Image: Red

● Lamp1 Id: 11
 Element name: GenElemInst_9
 Type of element: Lamp1
 Tab Order: default
 Static optimized: False
 Position
 X: 28
 Y: 503

Width: 70
Height: 70
Variable: PLC_PRG.bLamp_Fan
Image settings
Transparent: False
Transparent color: Blue
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 63
Y: 538
Animation duration: 0
Background
Image: Blue

- Lamp1 Id: 13
Element name: GenElemInst_10
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 28
Y: 588
Width: 70
Height: 70
Variable: PLC_PRG.bLamp_Conveyorb
Image settings
Transparent: False
Transparent color: Yellow
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 63
Y: 623
Animation duration: 0
Background
Image: Yellow
- Lamp1 Id: 15
Element name: GenElemInst_11
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 102
Y: 588
Width: 70
Height: 70
Variable: PLC_PRG.bLamp_ConveyorF
Image settings
Transparent: False
Transparent color: Yellow
Isotropic type: Isotropic
Horizontal alignment: Left

Vertical alignment: Top
Center
X: 137
Y: 623
Animation duration: 0
Background
Image: Yellow
● Lamp1 Id: 17
Element name: GenElemInst_12
Type of element: Lamp1
Tab Order: default
Static optimized: False
Position
X: 102
Y: 319
Width: 70
Height: 70
Variable: PLC_PRG.bHMI_productLocation
Image settings
Transparent: False
Transparent color: Green
Isotropic type: Isotropic
Horizontal alignment: Left
Vertical alignment: Top
Center
X: 137
Y: 354
Animation duration: 0
Background
Image: Green
■ Rectangle Id: 19
Element name: GenElemInst_14
Tab Order: default
Static optimized: False
Text ID: 887
Type of element: Rectangle
Position
X: 136
Y: 23
Width: 415
Height: 30
Angle: 0
Radius setting
Radius: From style
Value: 5
Center
X: 343
Y: 38
Colors
Normal state
Frame color
Transparency: 255
Fill color
Transparency: 255

Alarm state
Frame color
Transparency: 255
Fill color
Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Element-XYChart-Secondary-CursorColor
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Appearance
Line width: 1
Fill attributes: Filled
Line style: Solid
Texts
Text: %s
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]
Font color
Transparency: 255
Absolute movement
Use REAL values: False
Text variables
Text variable: PLC_PRG.sHMI_Message
Font variables
Size: , <pt>
Color variables
Toggle color: <toggle/tap variable>
Animation duration: 0
Input configuration
OnDialogClosed: Configure...
OnMouseClicked: Configure...
OnMouseDown: Configure...
OnMouseEnter: Configure...
OnMouseLeave: Configure...
OnMouseMove: Configure...
OnMouseUp: Configure...
OnValueChanged: Configure...
Toggle
Toggle on up if captured: False
Tap
Tap FALSE: False
Tap on enter if captured: False
■Rectangle Id: 21
Element name: GenElemInst_16
Tab Order: default
Static optimized: False
Text ID: 960
Type of element: Rectangle
Position

X: 142
Y: 73
Width: 245
Height: 30
Angle: 0
Radius setting
 Radius: From style
 Value: 5
Center
 X: 264
 Y: 88
Colors
 Normal state
 Frame color
 Transparency: 255
 Fill color
 Transparency: 255
 Alarm state
 Frame color
 Transparency: 255
 Fill color
 Transparency: 255
Use gradient color: True
Gradient setting
 Gradient
 Color1: Yellow
 Color2: Color [A=255, R=255, G=255, B=255]
 Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Appearance
 Line width: 1
 Fill attributes: Filled
 Line style: Solid
Texts
 Text: %d °C
Text properties
 Horizontal alignment: Centered
 Vertical alignment: Centered
 Text format: Default
 Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]
 Font color
 Transparency: 255
Absolute movement
 Use REAL values: False
Text variables
 Text variable: PLC_PRG.rHMI_targetTemp
Font variables
 Size: , <pt>
Color variables
 Toggle color: <toggle/tap variable>
Animation duration: 0
Input configuration
 OnDialogClosed: Configure...
 OnMouseClicked
 Input Action

- *Write Variable
 - Type: VisuDialogs.Numpad
 - Variable: <Text output variable>
 - Password: False
 - Dialog position type: UseGlobalSetting
 - OnMouseDown: Configure...
 - OnMouseEnter: Configure...
 - OnMouseLeave: Configure...
 - OnMouseMove: Configure...
 - OnMouseUp: Configure...
 - OnValueChanged: Configure...
 - Toggle
 - Toggle on up if captured: False
 - Tap
 - Tap FALSE: False
 - Tap on enter if captured: False
- Rectangle Id: 25
 - Element name: GenElemInst_19
 - Tab Order: default
 - Static optimized: False
 - Text ID: 263
 - Type of element: Rectangle
 - Position
 - X: 142
 - Y: 113
 - Width: 245
 - Height: 30
 - Angle: 0
 - Radius setting
 - Radius: From style
 - Value: 5
 - Center
 - X: 264
 - Y: 128
 - Colors
 - Normal state
 - Frame color
 - Transparency: 255
 - Fill color
 - Transparency: 255
 - Alarm state
 - Frame color
 - Transparency: 255
 - Fill color
 - Transparency: 255
 - Use gradient color: True
 - Gradient setting
 - Gradient
 - Color1: Yellow
 - Color2: Color [A=255, R=255, G=255, B=255]
 - Brightnesscolor: Color [A=255, R=0, G=0, B=0]
 - Appearance
 - Line width: 1
 - Fill attributes: Filled

Line style: Solid

Texts

Text: %.1f °C

Text properties

Horizontal alignment: Centered

Vertical alignment: Centered

Text format: Default

Font: Arial;12;Regular;Element-Background-Color, Color [A=255, R=255, G=255, B=255]

Font color

Transparency: 255

Absolute movement

Use REAL values: False

Text variables

Text variable: PLC_PRG.rHMI_corentTemp

Font variables

Size: , <pt>

Color variables

Toggle color: <toggle/tap variable>

Animation duration: 0

Input configuration

OnDialogClosed: Configure...

OnMouseClicked: Configure...

OnMouseDown: Configure...

OnMouseEnter: Configure...

OnMouseLeave: Configure...

OnMouseMove: Configure...

OnMouseUp: Configure...

OnValueChanged: Configure...

Toggle

Toggle on up if captured: False

Tap

Tap FALSE: False

Tap on enter if captured: False

Slider Id: 27

Element name: GenElemInst_21

Type of element: Slider

Tab Order: default

Static optimized: False

Position

X: 277

Y: 292

Width: 77

Height: 352

Variable: GVL.iRawAnalogIn

Move to click: False

Scale

Show scale: False

Scale start: 0

Scale end: 27648

Main scale: 10

Subscale: 5

Scale format (C syntax): %i

Scale proportion: 0

Bar

Diagram type: Bottom
Diagram type: Right
Orientation: Vertical
Running direction: Left to right
Running direction: Top to bottom
Center
X: 315
Y: 468
Animation duration: 0
Input configuration
OnValueChanged: Configure...
■ Button Id: 30
Element name: GenElemInst_22
Type of element: Button
Tab Order: default
Static optimized: False
Text ID: 749
Position
X: 28
Y: 193
Width: 98
Height: 75
Center
X: 77
Y: 230
Colors
Color
Transparency: 255
Alarm color
Transparency: 255
Use gradient color: True
Gradient setting
Gradient
Color1: Blue
Color2: Color [A=255, R=255, G=255, B=255]
Brightnesscolor: Color [A=255, R=0, G=0, B=0]
Button height: 0
Bitmap info
Scaling type: Isotropic
Transparent: False
Transparent color: Black
Horizontal alignment: Left
Vertical alignment: Top
Texts
Text: rest
Text properties
Horizontal alignment: Centered
Vertical alignment: Centered
Text format: Default
Font: Arial;28;Bold;Element-Button-FontColor, Color [A=255, R=0, G=0, B=0]
Font color
Transparency: 255
Font variables
Size: , <pt>

Button state variable
Digital variable: <NOT toggle/tap variable>

Animation duration: 0

Input configuration

OnDialogClosed: Configure...

OnMouseClicked: Configure...

OnMouseDown: Configure...

OnMouseEnter: Configure...

OnMouseLeave: Configure...

OnMouseMove: Configure...

OnMouseUp: Configure...

OnValueChanged: Configure...

Toggle

Toggle on up if captured: False

Tap

Variable: PLC_PRG.bHMI_reset

Tap FALSE: True

Tap on enter if captured: False

● Lamp1 Id: 32

Element name: GenElemInst_23

Type of element: Lamp1

Tab Order: default

Static optimized: False

Position

X: 153

Y: 198

Width: 70

Height: 70

Variable: PLC_PRG.bLamp_Reset

Image settings

Transparent: False

Transparent color: Blue

Isotropic type: Isotropic

Horizontal alignment: Left

Vertical alignment: Top

Center

X: 188

Y: 233

Animation duration: 0

Background

Image: Blue